DEPARTMENT OF MATHEMATICS, IIT Guwahati
MA102: Mathematics II
Tutorial Problems Sheet

21. Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be a basis for a vector space V. Show that the set $\{\mathbf{x}_1+\mathbf{x}_2, \mathbf{x}_2+\mathbf{x}_3, \dots, \mathbf{x}_n+\mathbf{x}_1\}$ is also a basis for V if and only if n is an odd integer.

Solution: We have

$$\begin{aligned} & a_1(\mathbf{x}_1+\mathbf{x}_2) + a_2(\mathbf{x}_2+\mathbf{x}_3) + \ldots + a_n(\mathbf{x}_n+\mathbf{x}_1) = \mathbf{0} \\ \Rightarrow\ & (a_1+a_n)\mathbf{x}_1 + (a_1+a_2)\mathbf{x}_2 + \ldots + (a_{n-1}+a_n)\mathbf{x}_n = \mathbf{0} \\ \Rightarrow\ & a_1+a_n = 0,\ a_1+a_2=0,\ \ \ldots,\ a_{n-1}+a_n = 0. \end{aligned}$$

The system of these equations has only the zero solution if and only if $\operatorname{rank}(A) = n$, where

$$A = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 & 1 \\ 1 & 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & 1 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & 1 & 0 \\ 0 & 0 & 0 & \ldots & 1 & 1 \end{bmatrix}.$$

If we apply the row operations $R_i \to R_i - R_{i-1}$ for $i = 2, 3, \dots, n$, one after the another on A, we arrive at a REF of A, in which the last row is $\begin{bmatrix} 0 & 0 & 0 & \ldots & 0 & 1+(-1)^{n-1} \end{bmatrix}$. Notice that $1+(-1)^{n-1} \neq 0$ iff n is odd, that is, $\operatorname{rank}(A) = n$ iff n is odd. Thus the set $S = \{\mathbf{x}_1+\mathbf{x}_2, \mathbf{x}_2+\mathbf{x}_3, \dots, \mathbf{x}_n+\mathbf{x}_1\}$ is linearly independent iff n is an odd integer. Since S contains n vectors, we conclude that S is a basis for V iff n is an odd integer. $\square$

22. Let W_1 and W_2 be two subspaces of a finite dimensional vector space V. Show that $\dim(W_1+W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$.

Solution: Choose a basis $S = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ for $W_1 \cap W_2$ and extend it to a basis $A = \{\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{y}_1, \dots, \mathbf{y}_p\}$ for W_1 and $B = \{\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{z}_1, \dots, \mathbf{z}_q\}$ for W_2. Thus $\dim(W_1) = n+p$ and $\dim(W_2) = n+q$. It is clear that $A \cup B$ spans W_1+W_2. Suppose $\sum\limits_{i=1}^{n} a_i\mathbf{x}_i + \sum\limits_{j=1}^{p} b_j\mathbf{y}_j + \sum\limits_{k=1}^{q} c_k\mathbf{z}_k = \mathbf{0}$. Then $\sum\limits_{k=1}^{q} c_k\mathbf{z}_k \in W_1 \cap W_2$ and therefore $\sum\limits_{k=1}^{q} c_k\mathbf{z}_k = \sum\limits_{i=1}^{n} d_i\mathbf{x}_i$ for some $d_i, i = 1, \dots, n$. Since B is a linearly independent set, $\sum\limits_{k=1}^{q} c_k\mathbf{z}_k = \sum\limits_{i=1}^{n} d_i\mathbf{x}_i$ implies that $c_k = 0$ for all k. Thus $\sum\limits_{i=1}^{n} a_i\mathbf{x}_i + \sum\limits_{j=1}^{p} b_j\mathbf{y}_j = \mathbf{0}$, and since A is a basis, we find that $b_j = 0$ for all j and $a_i = 0$ for all i. Thus $A \cup B$ is also linearly independent, and therefore $A \cup B$ is a basis for $W_1 + W_2$. Hence $\dim(W_1+W_2) = n+p+q = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$. $\square$

23. Let V be a finite dimensional vector space and let V_1 and V_2 be two subspaces of V. If $\dim(V_1+V_2) = \dim(V_1 \cap V_2) + 1$, show that either $V_1+V_2 = V_1, V_1 \cap V_2 = V_2$ or $V_1+V_2 = V_2, V_1 \cap V_2 = V_1$. [Equivalently, for subspaces V_1 and V_2 of V, if neither contains the other, then $\dim(V_1+V_2) \geq \dim(V_1 \cap V_2) + 2$].

Solution: Since $V_1 \cap V_2 \subseteq V_1 \subseteq V_1 + V_2$, we find that

$$\dim(V_1 \cap V_2) \leq \dim V_1 \leq \dim(V_1+V_2).$$

Thus the assumption $\dim(V_1 + V_2) = \dim(V_1 \cap V_2) + 1$ implies that either

$$\dim V_1 = \dim(V_1 \cap V_2) \text{ or } \dim V_1 = \dim(V_1 + V_2).$$

Now $\dim V_1 = \dim(V_1 \cap V_2)$ implies that $V_1 = V_1 \cap V_2$. Hence $V_1 \subseteq V_2$ and so $V_1 + V_2 = V_2$. Again, $\dim V_1 = \dim(V_1 + V_2)$ implies that $V_1 = V_1 + V_2$. Hence $V_2 \subseteq V_1$ and so $V_1 \cap V_2 = V_2$.

□

24. Find the coordinates of the vector $[1,2,3]^t$ with respect to the bases B and C for $\mathbb{R}^3$, where

$$B = \{[1,1,0]^t, [0,1,1]^t, [1,0,1]^t\} \text{ and } C = \{[1,1,1]^t, [1,1,-1]^t, [1,-1,1]^t\}.$$

Also, find the matrix P such that $[\mathbf{x}]_B = P[\mathbf{x}]_C$ for all $\mathbf{x} \in \mathbb{R}^3$.

Solution: Let $\mathbf{u} = [1,2,3]^t$. Solving $x[1,1,0]^t + y[0,1,1]^t + z[1,0,1]^t = [1,2,3]^t$, we get $x = 0, y = 2, z = 1$. Hence $[\mathbf{u}]_B = [0,2,1]^t$.

Solving $x[1,1,1]^t + y[1,1,-1]^t + z[1,-1,1]^t = [1,2,3]^t$, we get $x = \frac{5}{2}, y = -1, z = -\frac{1}{2}$. Hence $[\mathbf{u}]_C = [\frac{5}{2}, -1, -\frac{1}{2}]^t$.

For convenience, let $\mathbf{u}_1 = [1,1,1]^t, \mathbf{u}_2 = [1,1,-1]^t$ and $\mathbf{u}_3 = [1,-1,1]^t$. We have

$$[\mathbf{u}_1]_B = [\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]^t, [\mathbf{u}_2]_B = [\frac{3}{2}, -\frac{1}{2}, -\frac{1}{2}]^t \text{ and } [\mathbf{u}_3]_B = [-\frac{1}{2}, -\frac{1}{2}, \frac{3}{2}]^t.$$

Hence the required matrix P is given by

$$P = P_{B \leftarrow C} = [[\mathbf{u}_1]_B, [\mathbf{u}_2]_B, [\mathbf{u}_3]_B] = \frac{1}{2} \begin{bmatrix} 1 & 3 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & 3 \end{bmatrix}.$$

□